

The Fourier Transform

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1 The Fourier Transform

Let $f(\mathbf{x})$ be absolutely integrable on $\mathbb{R}^n$. Its **Fourier transform** $(\mathcal{F}f)(\boldsymbol{\xi})$ is defined by

$$(\mathcal{F}f)(\boldsymbol{\xi}) = \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \langle \boldsymbol{\xi}, \mathbf{x} \rangle} d\mathbf{x},$$

for $\boldsymbol{\xi} \in \mathbb{R}^n$. It is absolutely convergent precisely because $f(\mathbf{x})$ is absolutely integrable on $\mathbb{R}^n$. Let us first demonstrate some properties of the Fourier transform.

Proposition 1.1. *Let $f(\mathbf{x})$ and $g(\mathbf{x})$ be absolutely integrable on $\mathbb{R}^n$. Then the following properties hold:*

(i) *For any $\alpha, \beta \in \mathbb{R}$, we have*

$$(\mathcal{F}(\alpha f + \beta g))(\boldsymbol{\xi}) = \alpha(\mathcal{F}f)(\boldsymbol{\xi}) + \beta(\mathcal{F}g)(\boldsymbol{\xi}).$$

(ii) *If $g(\mathbf{x}) = f(\mathbf{x} + \boldsymbol{\alpha})$ for any $\boldsymbol{\alpha} \in \mathbb{R}^n$ then*

$$(\mathcal{F}g)(\boldsymbol{\xi}) = e^{2\pi i \langle \boldsymbol{\alpha}, \boldsymbol{\xi} \rangle} (\mathcal{F}f)(\boldsymbol{\xi}).$$

(iii) *If $g(\mathbf{x}) = f(A\mathbf{x})$ for any $A \in \text{GL}_n(\mathbb{R})$ then*

$$(\mathcal{F}g)(\boldsymbol{\xi}) = \frac{1}{|\det(A)|} (\mathcal{F}f)((A^{-1})^t \boldsymbol{\xi}).$$

Proof. Property (i) is immediate from linearity of the integral while property (ii) follows by applying the change of variables $\mathbf{x} \mapsto \mathbf{x} - \boldsymbol{\alpha}$. Property (iii) is proved by performing the change of variables $\mathbf{x} \mapsto A^{-1}\mathbf{x}$ which has Jacobian matrix A^{-1} with Jacobian determinant $\frac{1}{|\det(A)|}$. $\square$

The Fourier transform is intimately related to periodic functions. Let Λ be a complete integral lattice in $\mathbb{R}^n$ with fundamental domain $\mathcal{M}$ and denote the dual integral lattice by $\Lambda^\vee$. Suppose $f(\mathbf{x})$ is Λ -periodic and integrable on $\mathcal{M}$. Then we define the $\lambda^\vee$ -th **Fourier coefficient** $\hat{f}(\lambda^\vee)$ of $f(\mathbf{x})$ by

$$\hat{f}(\lambda^\vee) = \int_{\mathcal{M}} f(\mathbf{x}) e^{-2\pi i \langle \lambda^\vee, \mathbf{x} \rangle} d\mathbf{x}.$$

The **Fourier series** $S_f(x)$ is defined by

$$S_f(\mathbf{x}) = \frac{1}{V_\Lambda} \sum_{\lambda^\vee \in \Lambda^\vee} \hat{f}(\lambda^\vee) e^{2\pi i \langle \lambda^\vee, \mathbf{x} \rangle}.$$

Notice that the Fourier coefficients are only defined for elements of the dual integral lattice and that the Fourier series is a sum over these elements. A crucial question is where the Fourier series of $f(\mathbf{x})$ converges, if at all, and if so does it even converge to $f(\mathbf{x})$ itself. Under quite reasonable conditions the Fourier series converges uniformly to the function itself, but we first make a reduction. Fix a basis $\lambda_1, \dots, \lambda_n$ for Λ and let P be the associated generator matrix. This means P is the base change matrix from the standard basis to $\lambda_1, \dots, \lambda_n$ so that P is the unique invertible linear transformation on $\mathbb{R}^n$ satisfying $\Lambda = P\mathbb{Z}^n$ and $\Lambda^\vee = (P^{-1})^t \mathbb{Z}^n$. Letting $f_P(\mathbf{x}) = f(P\mathbf{x})$, we see that $f_P(\mathbf{x})$ is 1-periodic in each component and integrable on $[0, 1]^n$. As $\mathbb{Z}^n$ is self-dual and its covolume is 1, the $\mathbf{n}$ -th Fourier coefficient $\hat{f}_P(\mathbf{n})$ of $f_P(\mathbf{x})$ is given by

$$\hat{f}_P(\mathbf{n}) = \int_{[0,1]^n} f(\mathbf{x}) e^{-2\pi i \langle \mathbf{n}, \mathbf{x} \rangle} d\mathbf{x}.$$

and the Fourier series is

$$S_{f_P}(\mathbf{x}) = \sum_{\mathbf{n} \in \mathbb{Z}^n} \hat{f}_P(\mathbf{n}) e^{2\pi i \langle \mathbf{n}, \mathbf{x} \rangle}.$$

As P is invertible, it suffices to study the convergence properties of this Fourier series. The reduction here this that we may assume our Fourier series are 1-periodic in each component and in this case we have the following well-known result:

Theorem 1.2. *Suppose $f(\mathbf{x})$ is a smooth function on $\mathbb{R}^n$ and is 1-periodic in each component. Then the Fourier series of $f(\mathbf{x})$ converges uniformly everywhere to $f(\mathbf{x})$.*

In the case of 1-periodic functions of a single variable, we can do better as we may merely assume $f(x)$ is of bounded variation. This is known as the **Dirichlet-Jordan test**.

Theorem (Dirichlet-Jordan test). *Suppose $f(x)$ is a function on $\mathbb{R}$ which is 1-periodic and of bounded variation. Then the Fourier series of $f(x)$ converges locally uniformly to $f(x)$ on every set where $f(x)$ is continuous. Moreover, at any jump discontinuity, the Fourier series of $f(x)$ converges to the average of the left-hand and right-hand limits of $f(x)$. In particular, this holds for all continuously differentiable functions with at most a finite number of jump discontinuities.*